

HoverCanvas: Aerial Display Based on a Flying Screen and 100 m Long Range Dynamic Projection

Shimpei Yamamoto
Gunma University
Maebashi, Gunma, Japan
j251a026@gunma-u.ac.jp

Hiromasa Oku
Gunma University
Maebashi, Gunma, Japan
h.oku@gunma-u.ac.jp

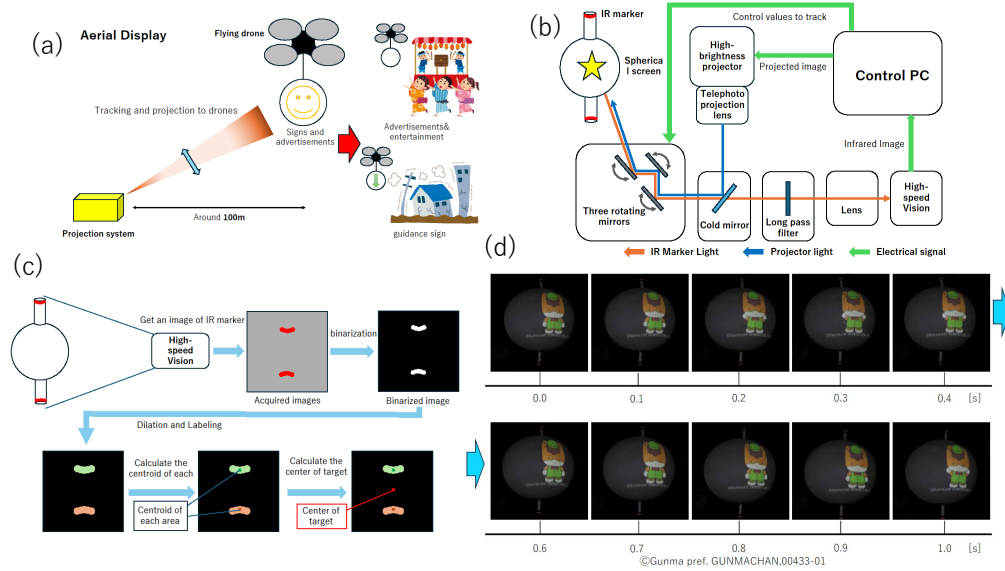


Figure 1: (a)Schematic figure of the proposed aerial display method named HoverCanvas, (b)Configuration of Hover Canvas system, (c)Summary of center of target estimation, (d)Sequential photographs of an animation being projected to match the position of a spherical screen 100 meters away.

Abstract

In recent years, technologies for creating aerial displays, such as drone light shows, have garnered significant attention. However, drone light shows have the disadvantage of requiring many drones and being expensive. To overcome this disadvantage, our group had proposed a cost-effective method for aerial displays using a spherical screen suspended by drones and a high-brightness projector. However, the previous methods faced several challenges in achieving large-scale, long-distance projections.

This study proposes a new method for projecting onto large spherical screens at long distances by developing a new infrared marker structure and a processing method. This method uses high-speed vision to detect the position of a moving screen using the developed IR marker and project images precisely at that position

to create an aerial display. The gaze direction of the high-speed camera and the projection direction of the projector was controlled with a high-speed optical gaze controller. The novel arrangement of the two IR light sources enabled dynamic projection onto larger screens at long distances.

To validate the proposed system, we conducted a field experiment. In this evaluation, a large spherical display with a diameter of 2.3 meters was suspended from a drone and maneuvered at a distance of 100 meters from the projection equipment. The system successfully demonstrated stable projection onto the moving screen.

CCS Concepts

• **Computing methodologies** → **Mixed / augmented reality**; • **Hardware** → **Displays and imagers**.

Keywords

aerial display, long range dynamic projection mapping, IR marker, high-speed gaze control

ACM Reference Format:

Shimpei Yamamoto and Hiromasa Oku. 2025. HoverCanvas: Aerial Display Based on a Flying Screen and 100 m Long Range Dynamic Projection. In

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA Posters '25, Hong Kong, Hong Kong

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2134-2/25/12

<https://doi.org/10.1145/3757374.3771457>

SIGGRAPH Asia 2025 Posters (SA Posters '25), December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3757374.3771457>

1 Introduction

Aerial displays, such as drone light shows, are gaining attention. However, their high cost limits widespread adoption. Research is actively exploring more cost-effective drone-based solutions[Hiroaki et al. 2011; Yamada 2018]. This study focuses on projecting images onto a drone-suspended screen, a promising alternative[Luchi et al. 2023].

Previous work by our group demonstrated dynamic projection onto a spherical screen using a long-range system with a telescopic three-mirror gaze control for high-speed vision and a telephoto projector[Hirohashi and Oku 2023, 2025]. However, this method faced challenges in large-scale, long-range applications, particularly concerning Infrared (IR) marker positioning and projection latency. This paper introduces HoverCanvas, a novel system that overcomes these limitations through a new IR marker design and a robust screen center estimation technique, enabling stable projection onto a large spherical screen from a long distance.

2 Hover Canvas system

The HoverCanvas system architecture is shown in Figure 1(b). It features a telescopic three-mirror gaze control system, based on the Saccade Mirror3[Iida and Oku 2016], which rapidly directs the gaze of a 1000fps high-speed vision system. This allows the system to track IR LED markers attached to the target screen. An IR long-pass filter is used to ensure stable detection at night by blocking visible light. A cold mirror aligns the projector's optical path with the vision system's gaze, allowing the projected image to follow the tracked markers without interference. A high-brightness projector with a telephoto lens ensures clear projection over long distances. A spherical screen is used, as its shape leverages the Coanda effect to enhance stability under the drone[Reynolds 1974].

3 Proposed Method

Previous methods, which used two closely positioned IR markers at the screen's top, suffered from three main issues in long-distance, large-scale projection[Hirohashi and Oku 2023]: (1) screen tilt caused self-occlusion of the markers; (2) markers were difficult to distinguish at a distance; and (3) significant projection lag occurred because positional corrections required slow projection image updates, rather than fast mirror control.

To overcome these limitations, our method utilizes two cylindrical IR markers placed at the poles (top and bottom) of the spherical screen. This polar arrangement offers two key advantages. First, it minimizes occlusion and maximizes separation, ensuring stable marker detection at long range. Second, it enables the elimination of latency-induced positional errors. The midpoint between the two markers accurately represents the screen's center. By tracking this midpoint with the high-speed mirror control for positional adjustments, we bypass the slower image update process for position correction. While rotation correction still requires image updates, its latency is less detrimental to overall stability. The cylindrical

markers, constructed from foam polystyrene and 1W IR LEDs, are attached to a 2.3-meter diameter screen.

The high-speed vision captures the IR emissions. Image processing involves binarization, morphological operations, and labeling to find the centroids of the two markers. The midpoint between these centroids is calculated as the screen center and is continuously tracked by the mirror control mechanism[Iida and Oku 2016], as summarized in Figure 1(c).

4 Evaluation Experiment

To validate the effectiveness of the proposed method, we conducted a field experiment. In this experiment, dynamic projections were performed on a large spherical screen (2.3 m diameter) suspended from a drone. The screen was maneuvered at a distance of 100 meters from the projection system to create dynamic conditions. We verified that the projection remained accurately centered on the screen without extending beyond its boundaries and appeared consistently bright and clear. The results in Figure 1(d) demonstrate that the projection remained centered on the moving screen without protruding beyond its edges, confirming the successful validation of the complete HoverCanvas system.

5 Conclusion

In this paper, we present HoverCanvas, a novel aerial display. Field experiments demonstrated stable, long-range (100m) dynamic projection onto a 2.3m screen, validating our polar IR marker tracking method for robust alignment on moving targets.

Acknowledgments

This work was partially supported by Japan Society for the Promotion of Science (JP24K02971, JP24K15843), JST Core Research for Evolutional Science and Technology (CREST) (JPMJCR23B2), JST-Mirai Program Grant Number JPMJMI24H1, Grant-in-Aid for the Priority Support Project (G3) at Gunma University in the fiscal year 2022-2024, and Grant-in-Aid for the Priority Support Project (G2) at Gunma University in the fiscal year 2025.

References

- Tobita Hiroaki, Maruyama Shigeaki, and Kuzi Takuya. 2011. Floating Avatar: Telepresence System Using Blimps for Communication and Entertainment (*CHI EA '11*). Association for Computing Machinery, New York, NY, USA, 541–550. doi:10.1145/1979742.1979625
- Yuito Hirohashi and Hiromasa Oku. 2023. Aerial Display Method Using a Flying Screen with an IR Marker and Long Range Dynamic Projection Mapping. In *SA'23: SIGGRAPH Asia 2023 Posters*. 1–2. doi:10.1145/3610542.3626123
- Yuito Hirohashi and Hiromasa Oku. 2025. Aerial display method using a flying screen and long range dynamic projection mapping. *IEEE Access (Volume: 13)* (2025). doi:10.1109/ACCESS.2025.3557000
- Kazuhisa Iida and Hiromasa Oku. 2016. Saccade Mirror 3: High-speed gaze controller with ultra wide gaze control range using triple rotational mirrors. In *2016 IEEE International Conference on Robotics and Automation (ICRA)*. 624–629. doi:10.1109/ICRA.2016.7487186
- Masatoshi Luchi, Yuito Hirohashi, and Hiromasa Oku. 2023. Proposal for an aerial display using dynamic projection mapping on a distant flying screen. In *2023 IEEE Conference Virtual Reality and 3D User Interfaces (VR)*. 603–608. doi:10.1109/VR55154.2023.00075
- A. J. Reynolds. 1974. *Turbulent flows in engineering*. John Wiley & Sons, Inc. 38–7–388 pages.
- Wataru Yamada. 2018. Spherical Drone Display. *The Journal of Image Information and Television Engineers* 72, 4 (2018), 504–507. doi:10.3169/itej.72.504